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Conversational Difference: perceptual differences between human-huamn and AI-generated dialogues

By

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Introduction

Since Alan Turing posed the famous question, "Can machines think?" in 1950, a new technology, Artificial Intelligence (AI), has emerged to simulate and expand human intelligence. (Jiang et al., 2024) In recent years, Artificial Intelligence (AI) has increasingly permeated everyone's everyday lives, transforming not only how we interact with technology but also how we communicate, absorb information, and make decisions. Among these developments, the world have witnessed a trend that large language model (LLM) based text-to-speech (TTS) emerges into the mainstream due to their high naturalness and zero-shot capacity. (Du et al., 2024) AI-driven communication systems, such as ChatGPT, Gemini, and Claude, have shown striking capabilities. These sophisticated models are not only capable of generating coherent, contextually rich dialogues but also of being fine-tuned to mimic specific individuals' voices and conversational styles with unprecedented fluency. Consequently, these technologies can enable numerous applications ranging from customer support and educational tools to entertainment platforms and journalism, manifesting prominently in chatbots, voice assistants, virtual influencers, and even fully AI-generated podcasts.

While the rapid growth and sophistication of AI-generated dialogue systems have widened numerous aspects of applications across various fields, this newly developed technology has also raise several concerns and critical debates among the public, including the ethics of the AI-generated conversations, data safety, and potential fraud that may occur due to the technological advancement. For instance, the noninvasiveness of voice data collection through accessible technology (such as smartphones, telehealth, and ambient recordings) or within clinical contexts means voice AI may help address health disparities and promote the inclusion of marginalized communities. However, the development of AI-ready voice datasets free from bias and discrimination is a complex

task.(Bélisle-Pipon et al., 2024) Beyond this, the authenticity, emotions, and dynamics of the AI-generated conversations have also been questioned. Historically, human conversations transcend mere informational exchange, serving instead as vital mediums for emotional connection, nuanced expression, and trust-building. As AI-generated conversations become increasingly indistinguishable from human interactions, it becomes essential to systematically investigate how audiences perceive these synthetic communications and whether they resonate similarly to authentic human dialogues.

In the past few years, researchers have widely explored this new technological advancement, but mainly only focused on the technical aspects or application aspects. For instance, Oord et al. (2016) investigated the text-to-speech fidelity or emotion simulation capabilities. On the other hand, there are also some researchers trying to discover how much will emotional modulation, intonation, and conversational dynamics affect listener perceptions and engagement(Kreuk et al., 2022). However, among these studies, a clear and significant gap still exists: few studies systematically bridge these technical advancements with empirical insights from communication research to explore real-world listener perceptions of structured spoken AI dialogues. Moreover, few studies engaged any structured and comparative contexts such as podcasts.

This thesis aims to fill the gap between the technical aspect and conversational aspect, and investigates and proves that humans' perception of the AI-generated conversations is also critical. In the past few years, limited research has been trying to discover the perceptual differences between AI-generated conversations (AI-AI dialogues) and human-human interactions. Previous work in this area mainly focused on informal contexts or customer service interactions. However, the increasing presence of autonomous AI-driven conversations has a significant impact on traditional interactions and thus requires more rig-

orous exploration: How do listeners perceive these AI-generated conversations compared to pure human conversations?

To achieve this goal, this study incorporates both quantitative and qualitative methods, trying to comprehensively analyze how listeners perceive and evaluate AI-generated conversations versus human conversations. Subjective ratings such as engagement, trust, and naturalness, and objective ratings, including language features, paralinguistic variables, tone and engagement, and conversation dynamics will be both analyzed. This research aims to seek meaningful and significant perceptual differences between these two types of conversations. Moreover, qualitative analysis, such as word frequency will be integrated to facilitate the understandings of the results.

Ultimately, with the combination of computational social science methodologies, well-established theory from the past, and human-centered principles, this study aims to contribute to multiple disciplines, including Human-Computer Interaction (HCI), media studies, and AI ethics. The results obtained from the study will shed some light on future improvement of AI communication technologies and will be able to inform the public audience on the ethical and psychological impacts of synthetic speech in an AI-prevalent world.

Literature Review

With the fast-evolving and the level of prevalence of generative AI, the intersection of artificial intelligence and human communication has drawn considerable attentions from researchers. Existing literature on AI-generated communication can be broadly categorized into three main areas: technical advancements in speech generation, perceptual challenges concerning naturalness and trust, and comparative analyses of interaction dynamics between human-human and human-AI conversations. The findings from previous studies in these

domains elaborate promising developments and critical limitations, further signaling key research gaps that this study aims to fill.

Technical Advancement in AI-Generated Speech

Significant technical milestones have marked the evolution of AI-generated speech in recent years. One landmark development was WaveNet, introduced by Oord et al. (2016), which dramatically improved the text-to-speech (TTS) systems by synthesizing natural-sounding human speech through deep neural networks. This new advancement set the solid base for the following developments in speech authenticity and naturalness. Based on this, Kreuk et al. (2022) pushed the field further by designing a novel method of textless speech emotion conversion, indicating that AI has the potential for generating emotionally expressive speech without textual inputs. Moreover, Wang et al. (2023) developed a neural codec language model (VALL-E) using discrete codes derived from an off-the-shelf neural audio codec model and found that VALL-E emerges in-context learning capabilities and can be used to synthesize high-quality personalized speech with only a 3-second enrolled recording of an unseen speaker as an acoustic prompt.

Beyond how fast the model can react, researchers also found the consistency of the audio generated is also remarkable. Borsos et al. (2023) introduced AudioLM, a framework for high-quality audio generation with long-term consistency, which they noticed that by training on large corpora of raw audio waveforms, AudioLM learns to generate natural and co-herent continuations given short prompts.

Despite these astonishing findings, current AI-speech generation often prioritizes the clarity and understandability of the content but ignores many details in the conversations such as speech variability and conversational dynamics. Such subtleties have raised concerns about the social and emotional authenticity

of these types of speech, particularly in structured, context-rich conversational situations such as podcasts

Perceptual Challenges: Engagement, Emotion, and Trust

Beyond technical advancements, humans' perception of AI-generated speech still remains complicated. For instance, B. Zhang et al. (2025) noticed that even highly convincing deepfake audio will still be perceived as emotionally superficial, particularly in longer and more nuanced interactions. Moreover, their research indicated that people do not always consider the deepfake audio as reliable. This finding supports the "Word-of-Machine" effect proposed by Longoni and Cian (2020), which highlights a listener preference for human-generated content in scenarios requiring emotional connection and creative engagement, despite accepting AI-generated outputs in more analytical contexts.

While some research indicate the AI-generated speech are being perceived as insincere and questionable, others show the different findings. Diel and Lewis (2024) designed a study to investigate the uncanny valley effect, which is the negative evaluation of near humanlike artificial entities, caused by synthetic speech. Their results suggest that while voices can fall into an uncanny valley, synthetic voices successfully escape it. The results support the account that uncanniness is caused by deviations from familiar categories, rather than categorical ambiguity or the misattribution of mind or animacy.

Trust and perceived authenticity is another critical dimension that will affect listener perceptions. Y. Zhang et al. (2025) emphasizes that nonverbal elements, such as smile intensity and vocal warmth, significantly influence perceptions of trustworthiness and competence. Unfortunately, these factors are often missing or inadequate in the AI-generated speech. These findings demonstrate the critical perceptual limitations of AI-generated conversations, especially when direct human relational cues are insufficient.

Human-AI and Human-Human Interaction Dynamics

Since the first occurrence of the "thinking machine", AI technology has experienced rapid growth and advancement in the following decades, profoundly impacting on diverse industries and modifying societal structures. Due to the autonomous nature of the artificial intelligence, the interactions between human and AI is drastically different from traditional human-computer interaction. AI has evolved beyond being a mere tool and is gradually becoming a companion, partner, friend, and even an opponent. (Jiang et al. (2024))

Therefore, interaction dynamic is another crucial factors that can influence the perception of the quality of the conversations dramatically. However, this aspect of the conversation is also frequently overlooked when evaluating the conversation quality. Comparative studies between natural and AI-mediated interactions shed important insights. For instance, Hill et al. (2015) revealed that while human-chatbot interactions are typically longer, they feature simpler syntax and reduced lexical variety compared to human-human interactions. Similarly, Yeomans et al. (2023) stress the importance of paralinguistic factors, including intonation and conversational rhythm, in achieving conversational success and maintaining listener engagement.

However, existing literature typically examines human-AI interactions within less structured, often informal contexts, overlooking structured multi-agent AI interactions such as in the podcast settings. Consequently, human's perception of conversations in a well-structured, multi-agent, and dynamic settings remains largely unknown.

Identified Gaps and Study Contribution

Combining existing literature reveals a clear tension: technological development in AI-generated speech has advanced much faster than the understanding

of its social and perceptual impacts in fully synthetic conversational contexts. There are only limited empirical studies evaluating listener perceptions of AI-generated conversations in realistic and structured situations, such as podcasts. Additionally, few studies simultaneously examine linguistic, paralinguistic, behavioral dynamism, and emotional dimensions to provide an integrated understanding of listener evaluations.

This study will implement a controlled, well-designed study, and analyze the perceptual difference between AI-generated conversations and pure human conversations. The study will collect participants' ratings on various aspects such as language features, emotional nuances, perceived trust and naturalness in order to thoroughly understand how synthetic speech influence human perception. The findings generated from this study will be able to enrich theoretical insights, methodological approaches, and practical implications concerning AI-driven communication in human-computer interaction.

Methodology

Research Design

This research employs the within-subjects design, which will allow participants to experience both conversation types and rate each conversation type separately. This design will facilitate a more direct and clear comparison between the ratings for each conversation type. In order to reduce the order effect, such as fatigue, priming biases, and novelty bias, participants will be randomly assigned to two counterbalanced groups. The first group will start with human-human conversations followed by AI-generated conversations, while the other group will experience the reverse order. This design will ensure the observed perceptual differences in the results are valid and reliable.

The within-subject design is particularly advantageous in perceptual research, reducing inter-subject variability and enhancing sensitivity to subtle differences in participant responses. This aligns effectively with the study's goal to isolate specific linguistic, acoustic, and paralinguistic features contributing to perceptions of conversational authenticity, coherence, engagement, and trustworthiness.

With the implementation of the within-subject design, we will be able to minimize the inter-subject variability, since different individuals may have different perceptions on the same language feature, which will influence the accuracy of the final results.

Stimuli Development

The experimental stimuli contain two dialogue types: human-human conversations and AI-generated dialogues. Human dialogues are selected from ten publicly available podcasts online. In order to avoid unfamiliarity, the content of the podcasts are all casual and common topics such as free diving, breathing techniques, and academic integrity. Each excerpt is edited to standardized durations ranging from 30 seconds to one minute. Moreover, each excerpt will have similar conversation flows, number of characters, and turn-taking frequency. This will make sure that participants' rating will not be affected by the assignment of excerpts.

AI-generated dialogues are developed using the Gemini Notebook Large Language Model (LLM), trained on the original human-human podcast transcripts to simulate the original human conversations. To further ensure comparability with human dialogues, each AI-generated conversation is structurally and semantically matched to corresponding human-generated excerpts, making sure that these two types of conversations will not have any drastic differences beyond the origin.

Standardizing the stimuli in both conversation types can ensure the comparability of the results obtained from each condition and thus improve the internal validity and facilitate a more reliable conclution about listener perception differences.

Participant Recruitment

The 70 participants in this study were recruited via the Prolific research platform with a medium effect size (Cohen's d = 0.5), alpha = 0.05, and statistical power of 0.80. The age range of these participants is from 19 to 81 and all the participants in the study are fluent in the experimental stimulus language (English). Participants who have hearing impairments or have been exposed to the study will be excluded. Demographic data, including age, gender, and ethnicity, are systematically collected for exploratory covariate analysis, ensuring sample representativeness and demographic balance. Ethical compensation guidelines outlined by Prolific are adhered to, and the study protocol has undergone IRB approval to ensure ethical compliance.

Measurement Instruments

Listener perceptions are captured using structured survey instruments administered immediately after each audio exposure. Key perceptual dimensions include Linguistic Diversity(LD), Tone and Engagement(TE), Paralinguistic Variables(PV) (e.g., filler words, intonation variability), and Behavioral Dynamic(BD) (e.g., smoothness and turn-taking behaviors). Survey items utilize five-point Likert scales to quantify these dimensions, enabling detailed quantitative comparisons between dialogue conditions.

For each perspective, participants will be asked by carefully designed questions. For instance, to measure the LD, participants will be asked to answer questions like "How often did the conversation contain emotional expressions?"

and they will select the answer from very often to rarely. For TE, participants will be asked to evaluate "How would you rate the positivity of the tone?", the answer varies from very positive to very negative. In order to measure the PV, the study designed questions like "How often did the speakers/speaker pause unnaturally during their speech?" to see how frequent participants perceive the unnatural pause during the conversations. Finally, for the BD, participants will be asked to assess "How smooth was the turn-taking in the conversation?" and the answer will range from very smooth to very disjointed. Each answer will be labeled ,and transformed to numeric values for further analysis.

Trustworthiness, naturalness, and participant preferences are assessed explicitly through additional forced-choice items and open-ended qualitative responses(e.g. "Which type of conversation felt more natural to you?" and "Which conversation type seemed more engaging to you?"), providing a comprehensive understanding of subjective listener experiences.

Experimental Procedure

The informed consent will be signed by the participants before the start of the study. Once obtained consent from the participants, they will go through a brief demographic survey, asking about their age, race, and gender. Subsequently, participants will be randomly assigned to one of the two counterbalanced conditions outlined in the previous section. After listening to the conversations (human or AI conversations), participants will be asked to complete a rating survey asking about their perception on the LD, TE, PV, and BD. After completing both conditions, participants will respond to some open-ended questions asking about their preference of the conversation types and their perception of the overall naturalness of each type of conversation. The experiment ends with a standard debriefing session explaining the nature of AI-generated dialogues and clarifying the study's objectives.

Data Analysis

To fully understand the data obtained from the experiment, this study incorporates statistical hypothesis testing, machine learning techniques, and qualitative analysis in the data analysis. Statistical analyses include paired-sample t-tests for rating comparisons, chi-square tests for further evaluating categorical preference data, and logistic regression for identifying influential predictors of listener preferences. Qualitative data from open-ended responses will be quantified by using word frequency and sentiment analysis, displaying the results using the word cloud, which enhances interpretability and complementing quantitative findings.

By using the analytical approaches mentioned above, the results obtained should be solid and sophisticated, which will provide adequate information for the interpretations in the following sections.

Result

Descriptive Statistics

Participants being recruited in the study range from 19 to 81 years old, with a mean of 40.84, a median of 38, and a standard deviation of 15.07, indicating an age-balanced sample. Moreover, the interquartile range of the sample is between 28.25 and 51.75, indicating that the sample is generalizable in terms of the whole population.

As for gender distribution, 40 out of 70 participants identified themselves as male, 29 out of 70 as female, and 1 out of 70 as non-binary. Even though the sample is slightly biased toward the male population, it should not cause major effect on the final results.

Perceptual Rating Averages Across Conditions

Participants evaluated podcast conversations along four main perceptual dimensions: Linguistic Diversity (LD), Tone and Engagement (TE), Paralinguistic Variable (PV), and Behavioral Dynamics (BD). The higher the ratings, the more human-like the perception of the conversation is.

Results show that the mean of LD was rated at an average of 2.83 for human and 2.72 for AI; TE at 4.07 for human and 3.85 for AI; PV at 2.75 for human and 2.83 for AI; BD at 3.62 for human and 3.37 for AI. These statistics highlight that participants still perceive pure human conversations as more structured, engaging, and smoother than AI-generated conversations. The only dimension that AI-generated conversations are being rated higher is the paralinguistic variable aspect, suggesting that participants perceive the AI-generated conversations as having more natural filler words and pauses.

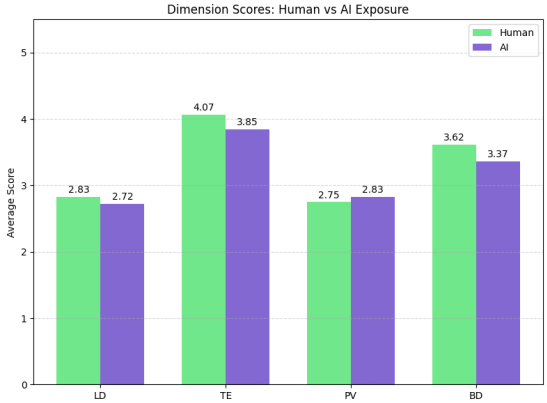

Figure 1: Average Ratings Across Conditions

Inferential Statistics: Paired t-tests on Exposure Average

Paired-sample t-test is implemented to analyze if the perceptual differences between AI-generated conversations and human conversations are statistically significant. For each dimension, the mean for human conversations will be

compared against its corresponding means for AI-generated conversations.

As shown in Table 1, the results indicate significant differences in three out of the four dimensions. For Linguistic Difference (LD), the difference was statistically significant ($p = 0.0079$), indicating that human conversations are perceived as more linguistically structured than AI-generated conversations. Similarly, in Tone and Engagement (TE), human dialogues were rated significantly higher ($p = 0.0067$), meaning that human conversations have more natural tones and are more engaging. For Behavioral Dynamics (BD), the t-test also reached significance ($p = 0.0166$), suggesting that participants found human conversations to feel smoother and more fluid in terms of dynamics

The only dimension that did not reach conventional levels of statistical significance is the Paralinguistic Variable (PV), which has a p-value of 0.0927. This result indicates that even though the mean of PV for AI-generated conversation is higher than human conversations, there is no significant difference between these two types of conversations.

These data confirm that even though AI-generated dialogues may have advanced in the past few decades and are capable of approaching human-like conversations, the perceptual gaps still exist, regardless of linguistic diversity, tone and engagement, or behavioral dynamics.

Dimension	**p-value**
Linguistic Differences	0.0079**
Tone and Engagement	0.0067**
Paralinguistic Variable	0.0927
Behavioral Dynamic	0.0166*

N = 70
Significance Levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 1: Paired t-tests comparing preferences between perceptual dimensions.

Preference Distribution

In addition to rating each conversation along perceptual dimensions, participants were asked to indicate their categorical preferences across three domains: perceived naturalness, host preference, and perceived engagement. These preference distributions will illustrate participants' subjective preferences.

For the question asking about the preferred naturalness ("Which type of conversation felt more natural to you?"), 33 participants (47%) selected human-human, 11 participants (16%) chose AI-AI, and 26 (37%) indicated both. This distribution highlights that nearly half of the respondents distinctly perceived human-generated speech as more natural, although a considerable portion of the sample either saw no difference or viewed both types favorably.

As for the preference question ("Which type of conversation would you prefer to listen to in real life?"), 42 participants (60%) chose a human-hosted podcast, while 15 out of 70 participants (21%) preferred an AI-hosted podcast. Thirteen participants (19%) selected "No preference", further suggesting that the majority of participants still prefer human-hosted podcasts, while there is still a considerable number of people who show no preference or prefer AI-

generated conversations.

A similar pattern emerged in response to the question "Which conversation type seemed more engaging to you?". Here, 27 participants (39%) preferred the human-human conversation, 16 (23%) chose AI, and 26 (37%) found both equally engaging. Only one participant indicated that neither conversation type was engaging. These engagement preferences suggest that while human dialogue is often seen as superior in affective terms, AI conversations are still capable of maintaining listener attention, especially when structured and voiced with care.

Together, these preference data suggest that while statistical perceptual differences between AI and human conversations are modest, participants' choices reflect clear social and emotional leaning toward human hosts and human-generated dialogue.

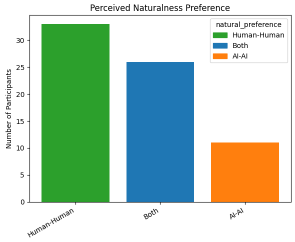

Figure 2: Preferred Naturalness

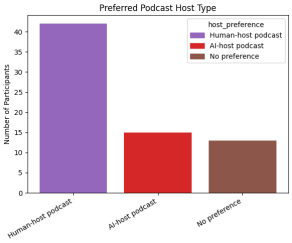

Figure 3: Preferred Host

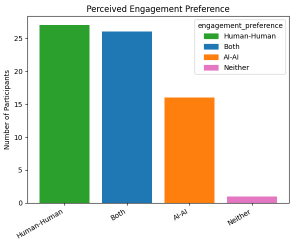

Figure 4: Preferred Engagement

Natural Preference by Gender and Age

To investigate whether gender or age played a significant role in shaping conversational preference, a chi-square test of independence was conducted using a contingency table of host preference (AI-AI, Human-Human, Both) by gender identity and age.

The chi-square test yielded a statistic of chi^2 = 1.7 for gender and chi^2 =

77.16 for age, with corresponding p-values of 0.791 and 0.441. Since none of the p-values exceed the significance threshold of 0.05, the result suggests that neither gender nor age was significantly associated with differences in natural conversation preference in this sample, further ensuring the reliability and validity of the results.

Predictive Modeling of Conversational Preference

To investigate the predictive utility of perceptual features more comprehensively, a logistic regression model was trained to predict participants' preferred conversation type (AI-AI, Human-Human, or Both) based on averaged scores across linguistic diversity, engagement, paralinguistic variables, and behavioral dynamics. The resulting classifier achieved an overall accuracy of 57% across the three classes, exceeding the chance baseline for a three-class prediction problem. The model demonstrated a macro-averaged F1-score of 0.61 and an ROC AUC score of 0.606, suggesting that the model is not perfectly predicting the performance but can provide some meaningful insights.

One interesting result shown in Table 2 is that the model predicted AI-AI preferences with high precision (1.00) but lower recall (0.67), while Human-Human preferences were detected with moderate balance (precision = 0.54, recall = 0.70). The "Both" category proved the most challenging, with a relatively low F1-score of 0.43. These results suggest that participants who strongly preferred either extreme (AI or human) exhibited clearer perceptual profiles, whereas ambivalent participants may reflect noisier or more context-dependent evaluations.

Overall, the model supports earlier findings that perceptual dimensions such as emotional expressiveness and trust will influence participants' preference. While not designed for deployment, this model provides a useful exploratory check on the consistency and structure of the perceptual data.

Condition	Precision	Recall	F1-score	support
AI-AI	1.00	0.67	0.80	3
Both	0.50	0.38	0.43	8
Human-Human	0.53	0.70	0.61	10
N = 70				

Table 2: Classification Report(Logistic Regression)

Word Frequency and Sentiment Analysis of Open-Ended Responses

Beyond quantitative analysis, qualitative open-ended responses were processed using text mining and natural language preprocessing (stopword removal, vectorization), using the wordcloud package from python. Word frequency analysis highlighted the most common adjectives across responses, with "engaging", "real", "natural", and "relatable" being the most frequent words. Word frequency analysis was also conducted for each response group.

As shown in the graphs presented below, for the participants who prefer human-human conversations, regardless of preference type, one of the most frequent key words in the answer is engaging. We can see that participants who prefer human conversations are more likely to think the conversations are realistic, natural, and have smoother interactions. Similar results are shown for participants who prefer AI-generated podcasts. We can see that participants also mentioned that the AI-AI conversations are feeling the flow of the conversations is good and the conversations are more informative, and smoother. Moreover, for the participants who show no preference between these two types of conversations, they also mentioned that the conversations of both types feel equally informative and relatable.

These qualitative insights indicate that regardless of people's preference,

engagement is one of the most important factors affecting people's perception of the conversations and this properties can be perceived from both types of the conversations.

Figure 5: Preferred Human

Figure 6: Preferred Both

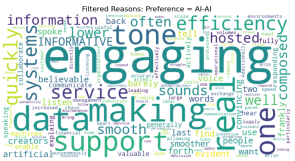

Figure 7: Preferred AI

Discussion

The findings from the study identify multiple perceptual differences between AI-generated and pure human conversations such as language features, pitch and tone, and behavioral dynamics. These discoveries are able to provide valuable theoretical insights and practical implications for the fast-developing artificial intelligence in the current society.

Interpretation of Key Findings

From the results presented in the data section, this study finds that even with the fast-evolving AI technology, humans still perceive significant differences between AI-generated conversations and pure human conversations. The differences listeners perceive in the language feature aspect indicate the importance of language structure in conversations and how dramatically it can shape people's perceptions. This finding is similar to prior research suggesting that listeners are sensitive to subtle linguistic cues that contribute to the naturalness and spontaneity of human speech (Hill et al., 2015; Yeomans et al., 2023).

For the differences in the aspect of tone and dynamism, they elaborate the importance of emotions and dynamism within the conversations, supporting prior studies emphasizing the centrality of emotional authenticity in listener engagement (Zhang et al., 2025; Cian & Longoni, 2020). On the other hand, the insignificant differences in the aspect of the paralinguistic variable also suggest that AI-generated conversations have already been able to simulate some aspect of human conversations without any perceptual differences.

Theoretical Contributions

These findings shed some light on both sides of the AI development and human-computer interaction(HCI). By empirically demonstrating specific perceptual strengths and limitations of AI-generated dialogues, the research enables us to comprehend that there are still some significant gaps between AI-generated conversations and pure human conversations, in terms of language features, tone, and behavioral dynamics. On the other hand, the study also demonstrates a wide acceptance and trust in AI-driven dialogues, informing a bright future for AI-mediated communication.

Practical Implications

Beyond the theoretical contributions, the practical implications of the study are also prominent. The findings in the study can offer valuable insights for AI model developers, informing the fact that despite the maturity of AI technology, several aspects of the conversations, such as language features and conversation dynamics, can be further improved to make the products more perceptually natural. These insights can facilitate the development of future artificial intelligence model that can be used to generate speech.

Moreover, users of these AI models can also benefit. For instance, content creators who use AI-generated conversations as part of their content can re-

examine the content and optimize user experiences by using a hybrid approach, using both AI and human dialogues, instead of fully AI-driven materials. This can not only potentially increase the profitability of their products but also improve the user experience.

Limitations and Future Research Directions

Despite rigorous methodology, this study has several limitations. First, the stimuli were limited to short podcast excerpts, and listener perceptions in longer, sustained conversations may differ significantly. Secondly, although a diverse participant sample was used, the sample size is relatively small to generate a generalizable results. Future studies should recruit more participants to further test the perception differences.

Beyond the sample size issue, future research can have multiple directions. In this study, participants were informed which dialogue is generated by AI at the beginning of the experiment, in order to reduce the effort of distinguishing the origin of the podcasts and to have a more direct comparison. However, it is reasonably to argue that revealing the origin of the podcast at the beginning can introduce some bias to the answers of the participants. When participants know the conversation is generated by AI but not pure human conversations, they may be more likely to rate the dialogues with lower ratings. Therefore, in the future, researchers can try not to reveal the origin of the dialogues at the beginning but let participants answer this question at the end of the experiment to see if the results have any significant changes.

Moreover, due to the time and resource limit of this research, the AI podcasts were all generated by the same model(Gemini Notebook LLM). This design will lack of the comparison of the stimuli generated by different models. In the future, the stimuli can be generated using different sophisticated models and randomly place them in the experiment. In the data analysis part, researchers

can try to analyze if the change the AI model will also change participants' perception of the dialogues.

If the future research can incorporate these changes, the results generated from the study should be much more comprehensive and have wider implications to the society.

Conclusion

In conclusion, this research offers robust empirical evidence illustrating clear perceptual differences preferring human-generated podcast conversations, particularly in terms of emotional depth, linguistic diversity, and behavioral dynamics. However, the potential and promise of AI-generated speech in achieving comparable paralinguistic realism provide encouraging directions for continued technological refinement. These findings significantly advance scholarly understanding of AI-mediated communication and provide actionable insights for both theoretical frameworks and practical implementations in media production and human-computer interaction.

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A Experiment Protocol

Objective

This study aims to investigate the perceptual differences between human-human conversations and AI-generated conversations in the podcast setting.

Hypothesis

The human-human conversations will be rated higher than the AI-generated conversations.

Participants

70 Participants being recruited in the study range from 19 to 81 years old, with a mean of 40.84, a median of 38, and a standard deviation of 15.07, the interquartile range of the sample is between 28.25 and 51.75. 40 out of 70 participants identified themselves as male, 29 out of 70 as female, and 1 out of 70 as non-binary.

Materials

Software: Python and Qualtrics

Procedure

- **Step 1:** Inform Consent
- **Step 2:** Participants are assigned to one of the two counterbalance conditions (Condition 1: AI-AI conversation first; Condition 2: Human-human conversation first)
- **Step 3:** Participants go through every conversations and answer the questions accordingly

- **Step 4:** Debrief

Variables

Dependent Variables:

- Linguistic Differences[LD]
- Tone and Engagement[TE]
- Paralinguistic Variables[PV]
- Behavioral Dynamics[BD]

Independent Variables

- Human-Human Conversations
- AI-AI Conversations

Data Collection

Data will be collected and exported via Qualtrics platform

Analysis Plan

All data anlysis will be conducted using Python

- Paired t-test (scipy package)
- chi-square test (panda package)
- Logistic Regression Model (scikit learn package)
- Word frequency and word cloud (wordcloud and nltk packages)

Ethical Considerations

All the participants in the study will sign informed consent before the beginning of the study, and all the procedures will follow the IRB protocols